Sets & set operations

Tools for describing collections — and for proving when two collections are equal.

Session	Date	Today
2 of 15	Tue 8 Sep 2026	Sets, operations, proof by double inclusion

RECAP

Where we left off

- → Rules come with conditions; check them before using a rule.
- → For inequalities, the sign tells you whether the direction stays or reverses.
- → One counterexample is enough to show an “always” claim is false.
- → Marks live in the justification, not the answer.

TODAY'S QUESTION

Yesterday we argued about *numbers*. But "the solution set of $|2x-3|<5$" is not a number — it is a **collection**.

How do you state, and *prove*, that two collections are the same?

OBJECTIVES

By the end of today you can...

- → Use **roster** and **set-builder** notation, and distinguish in from $subseteq$
- → Compute union, intersection, difference, complement, symmetric difference and Cartesian product
- → Prove set equalities by **double inclusion** and by element-chasing
- → Compute power sets and state why $|\mathcal{P}(A)| = 2^{|A|}$

> **MAIN SKILL** The third. The other objectives give us the vocabulary; double inclusion is the proof method we will practise carefully.

A set is a collection, and nothing more

DEFINITION

A **set** is an unordered collection of distinct objects, called its **elements**.

$x \in A$ means "x is an element of A";
$x \notin A$ means it is not.

TWO NOTATIONS

Roster — list them: $A = \{1, 2, 3\}$

Set-builder — state the rule:

$$B = \{\, x \in \mathbb{Z} : x^2 < 10 \,\}$$

read "the x in $\mathbb{Z}$ such that $x^2 < 10$", so $B = \{-3, -2, -1, 0, 1, 2, 3\}$.

UNORDERED, AND NO REPEATS $\{1, 2, 2, 3\}$ and $\{3, 1, 2\}$ are the **same set** as $\{1, 2, 3\}$. A set records *what* is in it, never how often or in what order.

The set with nothing in it

DEFINITION The **empty set** $\varnothing = \{\ \}$ has no elements: $x \in \varnothing$ is false for every x.

IT SHOWS UP ON ITS OWN $\{\, x \in \mathbb{R} : x^2 < 0 \,\} = \varnothing$.

Nobody chose this; the condition excludes everything.

TRAP · THREE DIFFERENT THINGS

$\varnothing$ — a set with no elements. $|\varnothing| = 0$.

$\{\varnothing\}$ — a set with **one** element, and that element is $\varnothing$. $|\{\varnothing\}| = 1$.

$\{0\}$ — a set with one element, the number zero. Not the same as either.

Element of, or subset of?

DEFINITION $A \subseteq B$ ("A is a **subset** of B") means: *every* element of A is an element of B.

$$A \subseteq B \iff \forall x\,(x \in A \Rightarrow x \in B)$$

THE DIFFERENCE IN ONE LINE $\in$ relates an **object** to a set. $\subseteq$ relates a **set** to a set.

EXAMPLE · $C = \{\{1\}, 2\}$

- $\{1\} \in C$ – true, it is listed.
- $\{1\} \subseteq C$ – **false**: is $1 \in C$? No.
- $1 \in C$ – false. C contains $\{1\}$, not 1.
- $\{2\} \subseteq C$ – true, since $2 \in C$.

Why $\varnothing \subseteq A$ for every set A

PROOF

To show $\varnothing \subseteq A$, we would need to find an element of $\varnothing$ that is not in A.

But $\varnothing$ has no elements. There is nothing to find.

So no element can break the subset condition, and $\varnothing \subseteq A$ for every set A. ■

WHY IT MATTERS This is why $\varnothing$ appears in every power set: it is a subset of every set.

TRAP $\varnothing \subseteq A$ always. But $\varnothing \in A$ only if you *put* it there.

CONCEPT I · EQUALITY

Two sets are equal when each contains the other

DEFINITION · SET EQUALITY

$$A = B \iff A \subseteq B \text{ and } B \subseteq A$$

Equivalently: A and B have exactly the same elements.

THE METHOD: DOUBLE INCLUSION

To prove $A = B$, write **two** proofs:

1. Take an arbitrary $x \in A$. Show $x \in B$. That gives $A \subseteq B$.
2. Take an arbitrary $x \in B$. Show $x \in A$. That gives $B \subseteq A$.

WHY THIS IS THE WHOLE SESSION The word **arbitrary** is what makes it a proof: you assumed nothing about x, so it holds for all of them.

ACTIVITY 1

True or false — and read it from the definition

ACTIVITY 1 PAIRS 10 MIN

Let $A = \{1, 2, \{3\}, \varnothing\}$. Decide true or false, and write the **one line of definition** that settles each.

1. $3 \in A$
2. $\{3\} \in A$
3. $\{3\} \subseteq A$
4. $\varnothing \in A$
5. $\varnothing \subseteq A$
6. $\{1, 2\} \subseteq A$
7. $|A| = 4$

Read the symbol, then read the definition

CLAIM		WHY
1 · $3 \in A$	False	A lists $\{3\}$, not 3.
2 · $\{3\} \in A$	True	It is listed.
3 · $\{3\} \subseteq A$	False	Would need $3 \in A$.
4 · $\varnothing \in A$	True	Only because it was listed.
5 · $\varnothing \subseteq A$	True	Vacuously, for *every* set.
6 · $\{1, 2\} \subseteq A$	True	Both are in A.
7 · $\lvert A \rvert = 4$	True	1, 2, $\{3\}$, $\varnothing$.

EXAMPLE · ROWS 2 AND 3 $\{3\} \in A$ asks if $\{3\}$ is *on the list*: yes. $\{3\} \subseteq A$ asks if its *contents* are: no.

Building new sets from old ones

UNION

$$A \cup B = \{\, x : x \in A \text{ or } x \in B \,\}$$

"or" is inclusive — x may be in both.

INTERSECTION

$$A \cap B = \{\, x : x \in A \text{ and } x \in B \,\}$$

Empty intersection ⇒ disjoint.

DIFFERENCE

$$A \setminus B = \{\, x : x \in A \text{ and } x \notin B \,\}$$

In A, not in B. Not symmetric.

WORKED · $A = \{1, 2, 3, 4\}$, $B = \{3, 4, 5\}$ $A \cup B = \{1, 2, 3, 4, 5\} \cdot A \cap B = \{3, 4\} \cdot A \setminus B = \{1, 2\} \cdot B \setminus A = \{5\}$

Complement and symmetric difference

COMPLEMENT

Fix a universe U. For $A \subseteq U$:

$$A^c = U \setminus A = \{\, x \in U : x \notin A \,\}$$

TRAP You must state U before using A^c. In $\mathbb{Z}$, the complement of the evens is the odds. In $\mathbb{R}$, it also contains non-integers such as $\frac{1}{2}$.

SYMMETRIC DIFFERENCE

$$A \triangle B = (A \setminus B) \cup (B \setminus A)$$

In exactly one of them — the "exclusive or" of sets.

WORKED With $U = \{1, \ldots, 6\}$, $A = \{1, 2, 3, 4\}$, $B = \{3, 4, 5\}$: $A^c = \{5, 6\}$, and $A \triangle B = \{1, 2\} \cup \{5\} = \{1, 2, 5\}$.

What a Venn diagram shows you

DEFINITION · VENN DIAGRAM A Venn diagram draws sets as overlapping regions — one region for each way an element can be in or out of each set.

EXAMPLE · TWO CIRCLES, FOUR REGIONS In both, in A only, in B only, in neither — that is $A \cap B$, $A \setminus B$, $B \setminus A$ and $(A \cup B)^c$. Every element of U lands in exactly one.

Translate: prose into notation, notation into prose

ACTIVITY 2 GROUPS OF THREE 12 MIN

Universe $U = \{1, 2, \ldots, 10\}$, $E = \{\text{even numbers in } U\}$, $T = \{\text{multiples of 3 in } U\}$.

1. Write E and T in roster notation, then in set-builder notation.
2. Compute $E \cap T$, $E \cup T$, $E \setminus T$, $E \triangle T$, T^c.
3. Turn into notation: "the numbers in U that are even but not multiples of three".
4. Turn into plain English: $(E \cup T)^c$.

Both directions of the translation

1–2 · THE SETS, AND THE OPERATIONS

$E = \{2, 4, 6, 8, 10\}$, $T = \{3, 6, 9\}$.

$E \cap T = \{6\} \cdot E \cup T = \{2, 3, 4, 6, 8, 9, 10\}$

$E \setminus T = \{2, 4, 8, 10\} \cdot E \triangle T = \{2, 3, 4, 8, 9, 10\}$

$T^c = \{1, 2, 4, 5, 7, 8, 10\}$

3 · PROSE INTO NOTATION "Even but not a multiple of three" is $E \setminus T = \{2, 4, 8, 10\}$.

4 · NOTATION INTO PROSE

$(E \cup T)^c$ is "neither even nor a multiple of three" — that is $\{1, 5, 7\}$.

Notice: *not (even or mult. of 3)* became *not even and not mult. of 3*. Hold that thought.

PART II

Proving that two sets are equal

Now use the vocabulary: prove a set equality one element at a time, or show a false identity with one clear example.

2

Proving an inclusion, one element at a time

THE METHOD

To show $X \subseteq Y$: take an **arbitrary** $x \in X$, unfold the definitions of the operations in X, and deduce $x \in Y$.

Each operation turns into a logical connective: $\cup \to$ or, $\cap \to$ and, $\setminus \to$ and-not.

THE SKELETON TO COPY

($\subseteq$) Let $x \in X$ be arbitrary. Then *[unfold]*, so *[deduce]*, hence $x \in Y$.

($\supseteq$) Let $x \in Y$ be arbitrary. Then *[unfold]*, so *[deduce]*, hence $x \in X$.

Both inclusions hold, so $X = Y$. $\blacksquare$

$A \setminus (B \cap C) = (A \setminus B) \cup (A \setminus C)$

($\subseteq$) — LEFT INTO RIGHT

Let $x \in A \setminus (B \cap C)$ be arbitrary.

Then $x \in A$ and $x \notin B \cap C$.

Now $x \notin B \cap C$ means **not** $(x \in B$ and $x \in C)$ – so $x \notin B$ or $x \notin C$.

If $x \notin B$, then $x \in A \setminus B$; if $x \notin C$, then $x \in A \setminus C$. Either way x lies in the union. ∎

THE STEP TO WATCH Negating "and" produced "or". That single move is De Morgan — two slides away, with a name.

...and the inclusion back again

($\supseteq$) – RIGHT INTO LEFT

Let $x \in (A \setminus B) \cup (A \setminus C)$ be arbitrary, so x is in at least one part.

Case 1: $x \in A \setminus B$. Then $x \in A$, $x \notin B$, so $x \notin B \cap C$.

Case 2: $x \in A \setminus C$. Then $x \in A$, $x \notin C$, so $x \notin B \cap C$.

Both give $x \in A \setminus (B \cap C)$. With the first half: equal. $\blacksquare$

WHY TWO CASES "Or" hands you one of two situations, not a specific one. Handle each, and both must land in the same place.

De Morgan's laws

THEOREM · DE MORGAN'S LAWS For all $A, B \subseteq U$:

$$(A \cup B)^c = A^c \cap B^c$$

$$(A \cap B)^c = A^c \cup B^c$$

PROOF OF THE FIRST

($\subseteq$) Let $x \in (A \cup B)^c$: then not ($x \in A$ or $x \in B$), so $x \notin A$ and $x \notin B$, i.e. $x \in A^c \cap B^c$.

($\supseteq$) Let $x \in A^c \cap B^c$: then x is in neither, so $x \notin A \cup B$. ■

IN ENGLISH "Not (even or a multiple of three)" $=$ "not even and not a multiple of three" — your Activity 2 sentence. Negation swaps *and* with *or*.

Showing a set identity is false

WHAT A DISPROOF NEEDS A set identity claims to work for **all** sets. To show it is false, give *one* example where the two sides differ.

WORKED · IS $A \setminus (B \setminus C) = (A \setminus B) \setminus C$?

Take $A = B = C = \{1\}$.

Left: $B \setminus C = \varnothing$, so $A \setminus \varnothing = \{1\}$.

Right: $A \setminus B = \varnothing$, so $\varnothing \setminus C = \varnothing$.
$\{1\} \neq \varnothing$: **false**.

TRAP

"It failed when I drew it" is not a disproof.

Name the sets, compute both sides, point at the element on one side and not the other — here, 1.

ACTIVITY 3

Prove it or break it

ACTIVITY 3 GROUPS OF THREE 15 MIN

Three are true and two are false. Prove a true one by double inclusion. For a false one, name the sets and compute both sides.

1. $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$
2. $(A \setminus B) \cup B = A \cup B$
3. $(A \cup B) \setminus B = A$
4. $A \triangle A = \varnothing$
5. $A \cup (B \cap C) = (A \cup B) \cap C$

Three survive, two do not

IDENTITY		PROOF SKETCH, OR THE WITNESS
1 · $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$	True	Distributivity: the "or" splits into two cases.
2 · $(A \setminus B) \cup B = A \cup B$	True	Remove B, put it back.
3 · $(A \cup B) \setminus B = A$	False	$A = B = \{1\}$: $\varnothing$ against $\{1\}$.
4 · $A \triangle A = \varnothing$	True	Nothing is in A but not in A.
5 · $A \cup (B \cap C) = (A \cup B) \cap C$	False	$A = \{1\}$, $B = C = \varnothing$: $\{1\}$ against $\varnothing$.

REPAIRING 3 It holds exactly when $A \cap B = \varnothing$.

The Cartesian product

DEFINITION · CARTESIAN PRODUCT The **Cartesian product** of A and B is

$$A \times B = \{\, (a,b) : a \in A \text{ and } b \in B \,\}$$

Its elements are **ordered pairs**: $(a,b) = (c,d)$ exactly when $a = c$ and $b = d$.

ORDERED, UNLIKE A SET $\{1,2\} = \{2,1\}$, but $(1,2) \neq (2,1)$ — the difference that makes coordinates, and tomorrow's functions, possible.

WORKED · $A = \{1,2\}$, $B = \{x,y\}$

$$A \times B = \{(1,x),(1,y),(2,x),(2,y)\}$$

$$B \times A = \{(x,1),(x,2),(y,1),(y,2)\}$$

TRAP $A \times B \neq B \times A$ in general. Witness: $A = \{1\}$, $B = \{2\}$.

How big is $A \times B$?

THEOREM

If A and B are finite, then $|A \times B| = |A| \cdot |B|$.

PROOF Every pair (a, b) comes from two independent choices: $|A|$ ways to pick a, then $|B|$ ways to pick b. Distinct choices give distinct pairs, since pairs match only when both coordinates do. So the count is $|A| \cdot |B|$. $\blacksquare$

YOU HAVE JUST USED THE PRODUCT RULE That argument — independent choices multiply — is the foundation of Session 7's combinatorics. Notice it now, so it is familiar when it gets a name.

The power set

DEFINITION · POWER SET The **power set** of A is the set of all subsets of A:

$$\mathcal{P}(A) = \{\, X : X \subseteq A \,\}$$

Its elements are themselves sets.

TRAP Members of $\mathcal{P}(A)$ are related to A by $\subseteq$, but to $\mathcal{P}(A)$ by $\in$. So $\{1\} \in \mathcal{P}(A)$ and $\{1\} \subseteq A$ – both true, different statements.

WORKED · $\mathcal{P}(\{1,2,3\})$

Organised by size:

$\varnothing$

$\{1\},\ \{2\},\ \{3\}$

$\{1,2\},\ \{1,3\},\ \{2,3\}$

$\{1,2,3\}$

Eight subsets: $1+3+3+1=8=2^3$. Both $\varnothing$ and A itself are always in there.

$|\mathcal{P}(A)| = 2^{|A|}$

THEOREM If $|A| = n$ then $\mathcal{P}(A)$ has exactly 2^n elements.

PROOF · ONE BINARY CHOICE PER ELEMENT

To build a subset $X \subseteq A$, walk the n elements of A and for each decide: in X, or not in X. Two options, n times, independently.

Every subset arises from exactly one such sequence of decisions, so there are $2 \times 2 \times \cdots \times 2 = 2^n$ of them. ■

CHECK IT $|A| = 0$: only $\varnothing$, and $2^0 = 1$.
$|A| = 3$: the eight we listed, $2^3 = 8$.

GROWS FAST $|A| = 10$ gives 1024 subsets; $|A| = 20$, over a million. Session 4 names this: exponential.

Products and power sets

ACTIVITY 4 PAIRS 12 MIN

Let $A = \{1, 2\}$ and $B = \{a\}$.

1. List $A \times B$ and $B \times A$. Are they equal? Justify from the definition of an ordered pair.
2. List $\mathcal{P}(A)$ in full. How many elements does $\mathcal{P}(\mathcal{P}(A))$ have? Do not list it.
3. Is $\mathcal{P}(A \cap B) = \mathcal{P}(A) \cap \mathcal{P}(B)$? Prove it or break it.
4. Is $\mathcal{P}(A \cup B) = \mathcal{P}(A) \cup \mathcal{P}(B)$? Prove it or break it.

Why ∩ survives and ∪ does not

1 · $A = \{1, 2\}$, $B = \{a\}$

$A \times B = \{(1, a), (2, a)\}$, $B \times A = \{(a, 1), (a, 2)\}$.

Not equal: $(1, a) = (a, 1)$ needs $1 = a$.

2 · POWER SETS GROW $\mathcal{P}(A) = \{\varnothing, \{1\}, \{2\}, \{1, 2\}\}$, so $|\mathcal{P}(\mathcal{P}(A))| = 2^4 = 16$.

3 · $\mathcal{P}(A \cap B) = \mathcal{P}(A) \cap \mathcal{P}(B)$ — TRUE $X \subseteq A \cap B$ means $X \subseteq A$ and $X \subseteq B$ — exactly the right-hand side.

4 · $\mathcal{P}(A \cup B) = \mathcal{P}(A) \cup \mathcal{P}(B)$ — FALSE With $A = \{1\}$, $B = \{2\}$: $\{1, 2\}$ is a subset of $A \cup B$ but of *neither* alone. Only $\supseteq$ holds.

WATCH OUT

Three mistakes worth avoiding

1 · CONFUSING $\in$ WITH $\subseteq$ Is $\{1\} \in \{\{1\}, 2\}$? Yes. Is $\{1\} \subseteq \{1, 2\}$? Yes. Different relations, different questions.

2 · THE DIAGRAM IS NOT THE PROOF A Venn diagram shows one configuration; the claim quantifies over all sets. Draw to guess, write the inclusion to prove.

3 · FORGETTING $\varnothing$ $\varnothing \subseteq A$ for every A. It is in every power set — and it is the case people drop when listing subsets.

A RELIABLE FIX Go back to the definition and read it as a sentence about an arbitrary element. These mistakes usually come from reasoning from the shape of the notation instead.

EXIT TICKET · 10 MINUTES

On your own, before you leave

1 With $U = \{1, \ldots, 8\}$, $A = \{2, 4, 6, 8\}$, $B = \{1, 2, 3, 4\}$: compute $A \triangle B$ and $(A \cap B)^c$.

2 Prove $(A \cap B) \subseteq A$ for all sets A, B. Two lines, and the word *arbitrary* must appear.

3 Disprove: $\mathcal{P}(A) \cup \mathcal{P}(B) = \mathcal{P}(A \cup B)$. Name your sets and the element that breaks it.

RULE FOR TODAY One sentence of justification under every answer. Hand it in on the way out — not graded, but it tells us who to sit with tomorrow.

WRAP-UP

What today was actually about

→ A set is determined by **what is in it** — not by order, repetition, or the notation you wrote it in.

→ Every operation is a logical connective in disguise: $\cup$ is *or*, $\cap$ is *and*, $\setminus$ is *and not*.

→ **Double inclusion** proves an equality; **one counterexample** destroys one.

set · element · subset · double inclusion · power set · De Morgan · Cartesian product · Venn diagram

Problem Set 2

6 identities to prove by double inclusion; **3 to disprove** by counterexample. Proofs begin with an arbitrary element. Disproofs name the sets and a separating element.

Next session: **Functions** — which are built from today's Cartesian product, and are the reason ordered pairs mattered.